

Facial Emotion Recognition

<https://github.com/aiyshwary/Facial-Emotion-Detection>

OVERVIEW

A comparative study of machine learning and deep learning approaches for recognizing facial emotions from images. The study benchmarks CNN, SVM, Random Forest, and Decision Tree models across multiple datasets.

IMPACT

Implemented and benchmarked four ML/DL models for 7-class facial emotion classification. CNN achieved approximately 75.5% accuracy, outperforming all traditional models on unseen data.

KEY DETAILS

1. Trained on icml_face_data and Kaggle FER datasets with 7 emotion classes (happy, sad, angry, fear, surprise, disgust, neutral).
2. CNN with data augmentation and stratified cross-validation achieved approximately 75.5% validation accuracy.
3. Random Forest reduced overfitting vs Decision Tree, achieving approximately 89.2% on ICML dataset.
4. SVM with PCA preprocessing achieved approximately 55% accuracy — lowest among all models on unseen data.
5. Concluded CNNs are best suited for FER due to hierarchical spatial feature extraction.
6. CNN architecture: 4 convolutional blocks (64-128-256-512 filters) with BatchNormalization, He Normalization, and Dense(512) to Dense(7, softmax) head totaling ~7.1M parameters.
7. Input images are 48x48 grayscale pixels normalized to [0,1]; augmentation via Keras ImageDataGenerator (rotation, width/height shift, shear, zoom (0.15), and horizontal flip).

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8. Training used Adam optimizer with EarlyStopping (patience=11), ReduceLROnPlateau (factor=0.5, min_lr=1e-6), 10-fold stratified cross-validation with batch size 32.